Energy 148 (2018) 1–15

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

A robust optimization approach for integrated community energy system in energy and ancillary service markets

Yizhou Zhou [a], Zhinong Wei [a, *], Guoqiang Sun [a], Kwok W. Cheung [b], Haixiang Zang [a], Sheng Chen [a]

[a] College of Energy and Electrical Engineering, Hohai University, Nanjing 210098, China
[b] GE Grid Solutions, Redmond, WA 98052, USA

ARTICLE INFO

Article history:
Received 6 June 2017
Received in revised form
12 January 2018
Accepted 14 January 2018
Available online 4 February 2018

Keywords:
Integrated community energy system
Uncertainty
Robust optimization
Energy market
Ancillary service market
Gaussian process

ABSTRACT

Distributed energy resources within local energy systems can be reorganized into a single entity, namely, into an integrated community energy system. This integration provides adequate scale to participate in wholesale markets. This paper proposes a day-ahead scheduling strategy for the integrated community energy system in a joint energy and ancillary service markets. The uncertainty of energy market prices, ancillary service market prices, wind power, and photovoltaic power are taken into account. The proposed integrated community energy system organizes combined cooling, heating, and power systems in different areas, and aggregates diverse distributed energy resources. Meanwhile, regulation up, regulation down, spinning, and non-spinning reserves are simultaneously employed in the proposed model. The robust optimization approach is adopted to handle uncertainty, and confidence intervals of uncertain parameters are predicted by a Gaussian process method. Finally, simulations of a real regional multi-energy system demonstrate the effectiveness and applicability of the proposed model.

1. Introduction

1.1. Integrated community energy system

Owing to an explosive increase of distributed energy resources, various heterogeneous coalitions have been implemented for energy integration at a local level, such as microgrids [1], virtual power plants (VPPs) [2], community energy systems [3], and combined cooling, heating, and power (CCHP) systems [4]. These local energy systems[1] have the advantages of energy utilization improvement, reliability enhancement and carbon emission reduction [5,6]. Nevertheless, local energy systems are often located in different regions and operated independently. In this regard, unreasonable resource allocation and insufficient flexibility are not negligible.

To address this issue, an integrated community energy system (ICES) has been proposed as an efficient approach to coordinate distributed energy resources and reorganize local energy systems [7,8]. The ICES captures attributes of the above-mentioned local energy systems and applies them to communities. The function of the ICES is not only to ensure energy requirements of local communities, but also to achieve energy exchanges and provide system services to the main grid.

From a development perspective, multi-energy carriers have become a trend in local energy systems [9,10]. The framework of local energy systems regarding multi-energy prosumer (producer–consumer) has been widely investigated in the literature. Xu et al. [11] developed a framework for the ICES to optimize interrelated thermal, gas, and electric power systems. Frameworks of the microgrid [12], the VPP [13], and the residential community energy system [14] have also been proposed to manage coordinated thermal and electrical schedules. A graphical paradigm of a combined distributed energy supply and CCHP system to provide cold/heat/electricity to customers was illustrated in Ref. [15].

* Corresponding author.
E-mail address: wzn_nj@263.net (Z. Wei).

[1] In this context, the heterogeneous coalitions for energy integration at the local level are regarded as local energy systems.

https://doi.org/10.1016/j.energy.2018.01.078
0360-5442/

1.2. Energy market and ancillary service market

ICES, as a single entity, has adequate scale to participate in wholesale markets with the integration of distributed energy resources [7]. Whereas scheduling strategy in an energy market (EM) has been extensively researched, the same problem in an joint EM and ancillary service market (ASM) is not that common [16,17].

Regulation, spinning, and non-spinning reserves are generally distinguished in most American ASMs [16]. Among these ancillary services, the spinning reserve represents the first consideration for the local energy system optimization in the joint day-ahead EM and ASM [18–21]. Shi et al. [18] proposed a stochastic bidding strategy for the microgrid, which consists of distributed generation units, renewable generation units, and battery energy storages. Based on the model proposed in Ref. [18], Zamani et al. [19] presented an optimization framework for day-ahead electrical and thermal energy resource scheduling in the VPP. A deterministic optimization algorithm was developed in Ref. [20] for a generation company with an integrated combined heat and power (CHP)-thermal-heat only system. Pousinho et al. [21] focused on the self-scheduling for a coordination of wind power plants with concentrated solar power plants. Meanwhile, optimization research in a joint energy and regulation markets has also been conducted in recent papers [22–24]. Day-ahead stochastic models were proposed in Refs. [22] and [23] for the VPP and the wind-thermal-pumped storage system, respectively. In addition, a scenario-based bidding strategy for a combined wind power and battery storage was presented in Ref. [24].

1.3. Robust optimization approach

The ICES participation in wholesale markets is subject to two main sources of uncertainty: 1) power output of renewable energy resources, and 2) market prices. Robust optimization (RO), which quantifies uncertainty with confidence intervals, has been considered as an alternative approach to stochastic programming for a lower computational burden and a less data dependence [25,26]. Furthermore, the RO approach could not only optimize for the worst realization, but also adjust the level of conservation by selecting different control parameters [27].

The RO approach has been sensitively employed for different energy system optimization problems. For example, RO models have been proposed to address uncertainty of wind power for unit commitment problem [28] and integrated energy system operation [29]. In addition, the uncertainty of market prices could be modeled via the RO approach, and hence, optimal offering curves were derived for the price-taker producer [30] and the VPP [31]. Taking both renewable power and EM price uncertainty into account, RO approaches were proposed for the CHP-microgrid in Ref. [32] and the VPP in Ref. [33].

1.4. Gaps in the literature and our contributions

Optimal problems of local energy systems participating in the joint EM and ASM are focused on either the spinning reserve or the regulation reserve. To the best of our knowledge, very few works discuss regulation, spinning and non-spinning reserves in a generic framework. Furthermore, the RO approach has not been adopted to address the uncertainty of ASM prices for local energy systems.

Based on the identified research gap, we propose a day-ahead scheduling strategy for the ICES in the joint EM and ASM, with the RO approach to handle the uncertainty of both EM and ASM prices, as well as of wind and photovoltaic power output. The ICES organizes CCHP systems in different areas, and integrates all three types of distributed energy resources: dispatchable generation units, renewable generation units and storage facilities. Instead of focusing on one type of ancillary service, we incorporate regulation up, regulation down, spinning, and non-spinning reserves into available ancillary services of ICES simultaneously. Moreover, Gaussian process (GP), an effective prediction intervals method, is employed to forecast the confidence intervals of uncertain parameters.

1.5. Paper organization

The rest of this paper is organized as follows. In Section 2, we describe the framework of the ICES and its market operation mechanisms. In Section 3, we provide the model formulation for the ICES in the joint of EM and ASM. In Section 4, we present a case study. Section 5 concludes the paper.

2. ICES framework and its market operation mechanisms

2.1. Framework of the ICES

The ICES is capable of effectively integrating local energy systems through various units, e.g., high-efficiency cogeneration (or trigeneration), renewable energy resources, and storage facilities [7,8]. In view of the fact that CCHP systems have a potential to significantly improve energy efficiency [34,35], in this work, CCHP systems are considered to be the organized local energy systems of ICES. As schematically depicted in Fig. 1, the structure of an CCHP-based ICES energy supply system incorporates four types of units: 1) renewable generation units (wind turbines and photovoltaic arrays), 2) dispatchable generation units (gas turbines and gas boilers), 3) storage facilities (electrical energy storages and thermal energy storages), and 4) energy delivery and transformation units (heat recovery systems, electric chillers, and absorption chillers). In this sense, primary energy sources, including natural gas, electricity, wind, and solar, are transformed into three different energy formats, i.e., electricity, heating, and cooling. We should clarify that, compared to current CCHP systems in Refs. [34,35], we further consider renewable generation units and storage facilities. This complies with a tendency for renewable generation units and storage facilities to be contained in local energy systems [36].

Meanwhile, the ICES can not only exchange energy between organized local energy systems but it can also trade energy with the main grid in wholesale markets [7]. As presented in Fig. 2, electricity, heating, and cooling energy can be exchanged through a radial electricity grid and a ring heating/cooling pipeline network. It should be noted that there is no direct electricity exchange between CCHP systems. The surplus electricity of each area can be sold to the main grid in the EM. Similarly, if energy demands are insufficient, the ICES can purchase electricity from the EM. The operation and control of the ICES are complemented with the applications of smart grid technologies [37]. As such, the economy and reliability of the overall system can be improved by the coordinated optimization of integrated units.

2.2. ICES operations in the EM and ASM

Without the loss of generality, the market structure described in this paper follows California electricity market mechanisms. Since the size of the considered ICES is not large enough, it functions as a price-taker agent in wholesale markets; namely, the behavior of the ICES would not affect the market prices. In addition, as a role of prosumer, the ICES can sell electricity for an hour and purchase it for another hour.

The ICES simultaneously bids in the EM and ASM 1 d in advance.

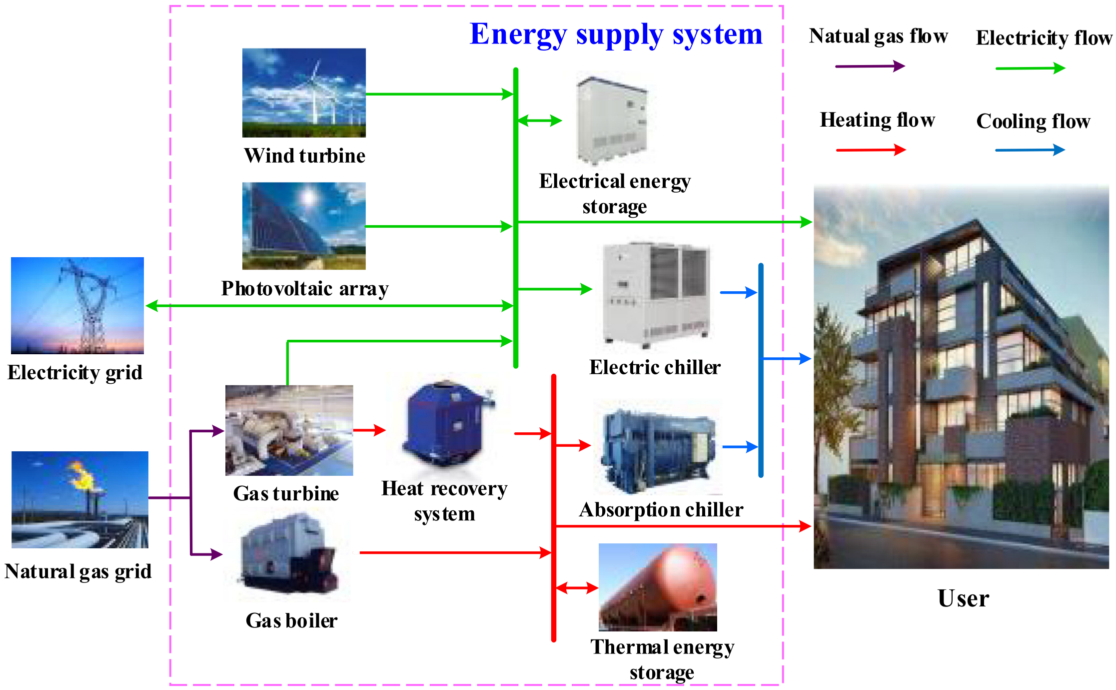


Fig. 1. Structure of the proposed ICES energy supply system with CCHP.

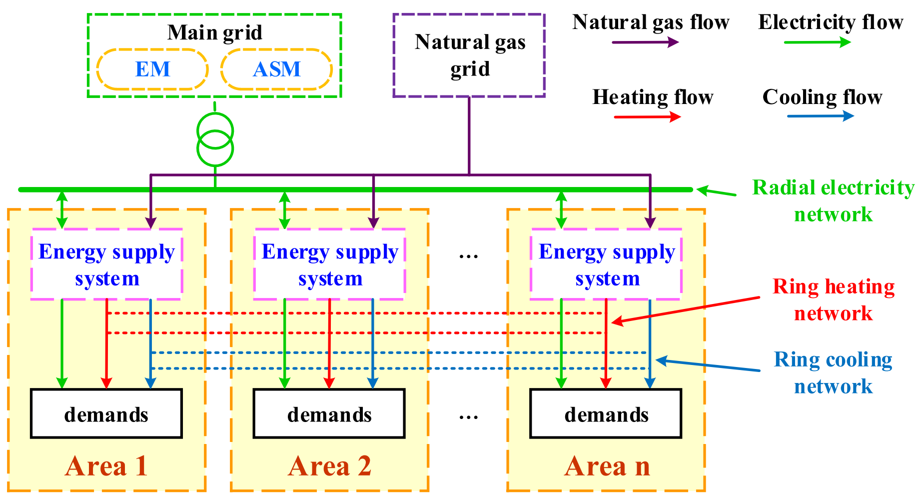


Fig. 2. Framework for an ICES integrating several CCHP systems.

After that, the California independent system operator (CAISO) accepts supply and demand curves from all of the market participants, and jointly clears EM and ASM prices for each hour of the scheduling day.

Four types of ancillary services are distinguished in California electricity markets: regulation up, regulation down, spinning, and non-spinning reserves [38]. The regulation up and regulation down are used to prevent energy imbalances and frequency fluctuations. In addition, both spinning and non-spinning reserves are prepared for energy balance restoration when a contingency occurs, but they are different in the requirement of a unit's status. The spinning reserve needs generation units already synchronizing with the grid, while the non-spinning reserve just requires generation units to start up rapidly. The full availability of committed reserve capacity should be delivered within a specified time period according to market rules [38,39].

3. Scheduling strategy for the ICES in the joint of EM and ASM

In this section, minimum total cost represents the objective of the ICES; this work does not consider the profit allocation problem. As a price-taker, the ICES optimizes its power traded in the EM and reserve capacities provided in the ASM for each hour of the next day.

It is worthy to emphasized that several binary variables are employed to descript the gas turbine status:

1) $\mu^{o}_{i,t}$ indicates whether a gas turbine is on for the power production and the reserve capacity provision. Note that four types of the above-mentioned reserves can be provided in this status

because the substitution of the regulation up or the spinning reserve for the non-spinning reserve is allowed.

2) $\mu_{i,t}^{sb}$ denotes whether a gas turbine is on standby to provide the non-spinning reserve, which is a special status between on ($\mu_{i,t}^{o}=1$) and off ($\mu_{i,t}^{o}=0$).

3) $\mu_{i,t}^{su}$ and $\mu_{i,t}^{sd}$ are auxiliary variables representing whether a gas turbine is startup and shutdown between 2 consecutive hours, respectively.

3.1. Deterministic model formulation

The proposed ICES participation in the joint EM and ASM is formulated as a mixed-integer linear programming (MILP) problem based on the expected values of market prices and renewable power in a deterministic model.

3.1.1. Objective function

$$\min\sum_{t=1}^{T}\sum_{i=1}^{n_i}\Bigg[-\left(\lambda_t^{e}P_{i,t}^{e}+\lambda_t^{ru}R_{i,t}^{ru}+\lambda_t^{rd}R_{i,t}^{rd}+\lambda_t^{sr}R_{i,t}^{sr}+\lambda_t^{nsr}R_{i,t}^{nsr}\right)+\left(\lambda_i^{f}\mu_{i,t}^{o}+\frac{860\lambda_t^{ng}}{HV^{ng}}\times\frac{P_{i,t}^{gt}}{\eta_i^{gt}}+\lambda_i^{sb}\mu_{i,t}^{sb}+\lambda_i^{su}\mu_{i,t}^{su}+\lambda_i^{sd}\mu_{i,t}^{sd}\right)+\left(\frac{860\lambda_t^{ng}}{HV^{ng}}\times\frac{P_{i,t}^{gb}}{\eta_i^{gb}}\right)\Bigg] \tag{1}$$

The objective function (1) consists of three terms. The first term denotes profit in electricity markets, incorporating the profit of energy sold/purchased in the EM and reserve capacities (regulation up, regulation down, spinning and non-spinning reserve capacities) provided in the ASM. The second term represents gas turbine costs, i.e., generation, standby, startup and shutdown costs. The generation cost is considered as a linear function of its electrical power production, and the standby cost is estimated as a fixed value in each standby hour. Meanwhile, each process of startup and shutdown operation is assumed to be a fixed cost. The third term defines the gas boiler cost. Note that the constant 860 in the second and third terms is the kWh-to-kcal ratio that achieves unit conversion of the natural gas price.

3.1.2. Constraints

1) Gas turbine constraints:

$$H_{i,t}^{hrs}=H_{i,t}^{gt}\eta_i^{hrs}=\frac{P_{i,t}^{gt}}{\eta_i^{gt}}\left(1-\eta_i^{gt}-\eta_i^{gtl}\right)\eta_i^{hrs}\quad\forall i,t \tag{2}$$

$$P_{i,t}^{gt}-R_{i,t}^{rd}\geq P_i^{gt,min}\mu_{i,t}^{o}\quad\forall i,t \tag{3}$$

$$P_{i,t}^{gt}+R_{i,t}^{ru}+R_{i,t}^{sr}\leq P_i^{gt,max}\mu_{i,t}^{o}\quad\forall i,t \tag{4}$$

$$P_{i,t}^{gt}+R_{i,t}^{ru}+R_{i,t}^{sr}+R_{i,t}^{nsr}\leq P_i^{gt,max}\left(\mu_{i,t}^{o}+\mu_{i,t}^{sb}\right)\quad\forall i,t \tag{5}$$

$$\left(P_{i,t}^{gt}+R_{i,t}^{ru}+R_{i,t}^{sr}+R_{i,t}^{nsr}\right)-\left(P_{i,t-1}^{gt}-R_{i,t-1}^{rd}\right)\leq r_i^{u}\quad\forall i,t \tag{6}$$

$$\left(P_{i,t-1}^{gt}+R_{i,t-1}^{ru}+R_{i,t-1}^{sr}+R_{i,t-1}^{nsr}\right)-\left(P_{i,t}^{gt}-R_{i,t}^{rd}\right)\leq r_i^{d}\quad\forall i,t \tag{7}$$

$$0\leq R_{i,t}^{ru}\leq T^{r}r_i^{u}\quad\forall i,t \tag{8}$$

$$0\leq R_{i,t}^{rd}\leq T^{r}r_i^{d}\quad\forall i,t \tag{9}$$

$$0\leq R_{i,t}^{ru}+R_{i,t}^{sr}+R_{i,t}^{nsr}\leq T^{s}r_i^{u}\quad\forall i,t \tag{10}$$

$$-\mu_{i,t}^{sd}\leq\mu_{i,t}^{o}-\mu_{i,t-1}^{o}\leq\mu_{i,t}^{su}\quad\forall i,t \tag{11}$$

$$\mu_{i,t}^{sb}\leq\left(1-\mu_{i,t}^{o}\right)\quad\forall i,t \tag{12}$$

$$T_i^{u,min}\mu_{i,t}^{su}\leq\sum_{h=t}^{t+T_i^{u,min}-1}\mu_{i,t}^{o}\quad\forall t\leq T-T_i^{u,min}+1,\forall i \tag{13}$$

$$T_i^{d,min}\mu_{i,t}^{sd}\leq\sum_{h=t}^{t+T_i^{d,min}-1}\left(1-\mu_{i,t}^{o}\right)\quad\forall t\leq T-T_i^{d,min}+1,\forall i \tag{14}$$

In (2), the thermal power production of heat recovery systems, $H_{i,t}^{hrs}$, is subjected to the by-product heat of gas turbines, $H_{i,t}^{gt}$. The power and reserve capacities provided by gas turbines should meet the lower and upper generation constraints in (3)–(5) and the ramp-up and ramp-down limits in (6) and (7). Moreover, reserve capacities should be limited by the ramping rate during the time for delivering reserves. Available regulation reserves are ensured by (8) and (9), while available spinning and non-spinning reserves are derived in (10). Constraints (11) and (12) are logical expressions to ensure the relationships of gas turbine binary variables. The minimum uptime and downtime constraints are presented in (13) and (14).

2) *Wind turbine and photovoltaic array constraints:* Equations (15) and (16) ensure wind and photovoltaic power output below available production (i.e., power spillage is considered):

$$P_{i,t}^{w}\leq P_{i,t}^{aw}\quad\forall i,t \tag{15}$$

$$P_{i,t}^{s}\leq P_{i,t}^{as}\quad\forall i,t \tag{16}$$

3) *Electrical and absorption chiller constraints:* In (17) and (18), the linear relationship between power input and power output of chillers is assumed by a simplified hypothesis, COP. The lower and upper power limits of chillers are addressed in (19) and (20). We have:

$$Q_{i,t}^{ec}=P_{i,t}^{ec}COP_i^{ec}\quad\forall i,t \tag{17}$$

$$Q_{i,t}^{ac}=H_{i,t}^{ac}COP_i^{ac}\quad\forall i,t \tag{18}$$

$$P_{i,t}^{ec,min}\leq P_{i,t}^{ec}\leq P_{i,t}^{ec,max}\quad\forall i,t \tag{19}$$

$$H_{i,t}^{ac,min}\leq H_{i,t}^{ac}\leq H_{i,t}^{ac,max}\quad\forall i,t \tag{20}$$

4) Electrical and thermal energy storage constraints:

$$S_{i,t}^{\mathrm{ees}} = S_{i,t-1}^{\mathrm{ees}} + \eta_i^{\mathrm{ce}} P_{i,t}^{\mathrm{ce}} - \frac{P_{i,t}^{\mathrm{de}}}{\eta_i^{\mathrm{de}}} \quad \forall i, t \tag{21}$$

$$0 \le P_{i,t}^{\mathrm{ce}} \le P_i^{\mathrm{ce,max}} \quad \forall i, t \tag{22}$$

$$0 \le P_{i,t}^{\mathrm{de}} \le P_i^{\mathrm{de,max}} \quad \forall i, t \tag{23}$$

$$S_{i,t}^{\mathrm{ees,min}} \le S_{i,t}^{\mathrm{ees}} \le S_{i,t}^{\mathrm{ees,max}} \quad \forall i, t \tag{24}$$

$$S_{i,0}^{\mathrm{ees}} = S_{i,24}^{\mathrm{ees}} = 0 \quad \forall i, t \tag{25}$$

$$S_{i,t}^{\mathrm{tes}} = S_{i,t-1}^{\mathrm{tes}} + \eta_i^{\mathrm{ct}} H_{i,t}^{\mathrm{ct}} - \frac{H_{i,t}^{\mathrm{dt}}}{\eta_i^{\mathrm{dt}}} \quad \forall i, t \tag{26}$$

$$0 \le H_{i,t}^{\mathrm{ct}} \le H_i^{\mathrm{ct,max}} \quad \forall i, t \tag{27}$$

$$0 \le H_{i,t}^{\mathrm{dt}} \le H_i^{\mathrm{dt,max}} \quad \forall i, t \tag{28}$$

$$S_{i,t}^{\mathrm{tes,min}} \le S_{i,t}^{\mathrm{tes}} \le S_{i,t}^{\mathrm{tes,max}} \quad \forall i, t \tag{29}$$

$$S_{i,0}^{\mathrm{tes}} = S_{i,24}^{\mathrm{tes}} = 0 \quad \forall i, t \tag{30}$$

Equation (21) presents the available electrical energy stored in the electrical energy storage at the end of each hour. Equations (22) and (23) denote the maximum charge and discharge power limits, while (24) imposes lower and upper limits on energy storage devices. The initial and final energy stored in the electrical energy storage are assumed to be 0, as in (25). In reference to the electrical energy storage constraints (21)–(25), the thermal energy storage constraints are given in (26)–(30).

5) Power balance constraints:

$$P_{i,t}^{\mathrm{gt}} + P_{i,t}^{\mathrm{w}} + P_{i,t}^{\mathrm{s}} + P_{i,t}^{\mathrm{de}} = P_{i,t}^{\mathrm{em}} + P_{i,t}^{\mathrm{el}} + P_{i,t}^{\mathrm{ec}} + P_{i,t}^{\mathrm{ce}} \quad \forall i, t \tag{31}$$

$$H_{i,t}^{\mathrm{hrs}} + H_{i,t}^{\mathrm{gb}} + H_{i,t}^{\mathrm{dt}} = H_{i,t}^{\mathrm{tl}} + H_{i,t}^{\mathrm{ac}} + H_{i,t}^{\mathrm{ct}} + \sum_{j \in n_i : j \ne i} H_{i,j,t}^{\mathrm{tf}} - \sum_{j \in n_i : j \ne i} H_{j,i,t}^{\mathrm{tf}} \left(1 - \eta^{\mathrm{tf}} d_{i,j}\right) \quad \forall i, t \tag{32}$$

$$Q_{i,t}^{\mathrm{ec}} + Q_{i,t}^{\mathrm{ac}} = Q_{i,t}^{\mathrm{cl}} + \sum_{j \in n_i : j \ne i} Q_{i,j,t}^{\mathrm{cf}} - \sum_{j \in n_i : j \ne i} Q_{j,i,t}^{\mathrm{cf}} \left(1 - \eta^{\mathrm{cf}} d_{i,j}\right) \quad \forall i, t \tag{33}$$

Equations (31)–(33) enforce the electrical, thermal and cooling power balance of injection power and withdrawal power in area i, respectively. In (31), the injection electricity includes the power production of the gas turbine, the wind turbine, and the photovoltaic array, as well as discharge power of the electrical energy storage. The withdrawal electricity comprises power traded in the EM, power demand for the electrical load and the electric chiller, as well as charge power of the electrical energy storage. In (32), the total heating production of the heat recovery system, the gas boiler, and the thermal energy storage should meet the total power demand for the absorption chiller and the heating load, as well as the thermal power flow between area i and other areas j. Equation (33) ensures the cooling power balance between the electric and absorption chiller power output, the cooling load demand, and the cooling power flow between area i and other areas j.

6) *Gas boiler and power flow constraints:* The gas boiler and thermal and cooling power flow between areas are subjected to the upper and lower operational limits:

$$\begin{cases} H_i^{gb,\mathrm{min}} \le H_{i,t}^{\mathrm{gb}} \le H_i^{gb,max} \\ 0 \le H_{i,j,t}^{\mathrm{tf}} \le H_{i,j}^{\mathrm{tf,max}} \\ 0 \le H_{i,j,t}^{\mathrm{cf}} \le H_{i,j}^{\mathrm{cf,max}} \end{cases} \quad \forall i, t \tag{34}$$

3.2. GP and uncertainty characterization

In this model, inputted confidence intervals of uncertain parameters are predicted by the GP [40], which has been applied in renewable power forecasting [41] and market prices forecasting [42], rather than assuming a single deviation of uncertain parameters. In this way, the completeness and accuracy of input parameters are guaranteed.

The uncertain parameters include: EM prices, ASM prices, available wind power output and available photovoltaic power output in the above-mentioned deterministic model. Furthermore, the uncertainty of wind and photovoltaic power output is mainly determined, and hence, can be described by the wind speed and solar irradiance uncertainty, respectively. Thus, the GP is employed to forecast confidence intervals of EM prices, ASM prices, wind speed, and solar irradiance.

The statistical features of the GP $f(x)$ can be completely indicated by its mean function $\mu(x)$ and covariance function $k(x, x')$:

$$f(x) \sim GP(\mu(x), k(x, x')), \tag{35}$$

where $\mu(x)$ and $k(x, x')$ are defined as:

$$\begin{cases} \mu(x) = E[f(x)] \\ k(x, x') = E[(f(x) - \mu(x))(f(x') - \mu(x'))] \end{cases}. \tag{36}$$

Given a training set $\boldsymbol{D}$=($\boldsymbol{X}$,$\boldsymbol{Y}$) with N input vectors, $\boldsymbol{X} = \{x_n\}_{n=1}^N$, and N corresponding targets, $\boldsymbol{Y} = \{y_n\}_{n=1}^N$, because the observed value y differs from the function value $f(x)$ by the associated noise ε, the relationships between input vectors and targets are satisfied:

$$y_n = f(x_n) + \varepsilon_n \quad \forall n, \tag{37}$$

where the noise ε_n is assumed to be $\varepsilon_n \sim N(0, \sigma_n^2)$. Note that constraints (37) comprise a linear relationship, and hence y forms a GP as follows:

$$y \sim GP\left(\mu(x), k(x, x') + \sigma_n^2 \delta_{mn}\right), \tag{38}$$

where δ_{mn} is the Kronecker delta. The covariance function can be expressed as a matrix $\boldsymbol{C}(\boldsymbol{X}, \boldsymbol{X})$ as:

$$\mathbf{C}(\mathbf{X}, \mathbf{X}) = \mathbf{K}(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}, \tag{39}$$

where $\boldsymbol{K}(\boldsymbol{X},\boldsymbol{X})$ is the kernel matrix with elements $\boldsymbol{K}_{mn} = k(x_m, x_n)$, and $\boldsymbol{I}$ is a unit matrix.

Since the training set is given, our goal is to obtain the predictive target f_* for a new input x_*. The joint Gaussian distributions of the predictive target f_* and the training set targets $\boldsymbol{Y}$ can be computed as follows:

$$\begin{Bmatrix} \mathbf{Y} \\ f_* \end{Bmatrix} \sim N\left\{0, \begin{bmatrix} \mathbf{K}(\mathbf{X},\mathbf{X})+\sigma_n^2\mathbf{I} & \mathbf{K}(\mathbf{X},x_*) \\ \mathbf{K}(x_*,\mathbf{X}) & k(x_*,x_*) \end{bmatrix}\right\}. \tag{40}$$

The prediction result f_* is also a Gaussian distribution, i.e., $f_*{\sim}N(\mu_*, \sigma_*^2)$, whose mean function μ_* and covariance function σ_*^2 are given by:

$$\begin{cases} \mu_* = \mathbf{K}(x_*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X})+\sigma_n^2\mathbf{I}\right]^{-1}\mathbf{Y} \\ \sigma_*^2 = k(x_*,x_*) - \mathbf{K}^{\mathrm{T}}(x_*,\mathbf{X})\left[\mathbf{K}(\mathbf{X},\mathbf{X})+\sigma_n^2\mathbf{I}\right]^{-1}\mathbf{K}(\mathbf{X},x_*) \end{cases}. \tag{41}$$

The Gaussian distribution can be conveniently converted into the confidence interval. Based on the predictive probability density in constraints (41), the α-level confidence interval of prediction result f_* becomes:

$$[Lb_{1-\alpha}(f_*), Ub_{1-\alpha}(f_*)] = \left[\mu_* - z_{(1-\alpha)/2}\sigma_*, \mu_* + z_{(1-\alpha)/2}\sigma_*\right], \tag{42}$$

where $Lb_{1-\alpha}(f_*)$ and $Ub_{1-\alpha}(f_*)$ are the lower and upper bounds of predictive target f_*, and α is the confidence level with a value within the interval [0,100%]. $z_{(1-\alpha)/2}$ is the critical value of standard normal distribution.

Afterwards, available wind power output can be calculated by wind speed according to the wind power conversion curve as follows:

$$P_{i,t}^{\mathrm{aw}} = \begin{cases} 0, & v_{i,t} < v_i^{\mathrm{ci}}, v_{i,t} > v_i^{\mathrm{co}} \\ P_i^{\mathrm{wr}}\left(a_3 v_{i,t}^3 + a_2 v_{i,t}^2 + a_1 v_{i,t} + a_0\right), & v_i^{\mathrm{ci}} < v_{i,t} < v_i^{\mathrm{r}} \\ P_i^{\mathrm{wr}}, & v_i^{\mathrm{r}} < v_{i,t} < v_i^{\mathrm{co}} \end{cases} \quad \forall i,t, \tag{43}$$

where P_i^{wr} is the rated power of the wind turbine (in kW). v_i^{ci}, v_i^{co}, and v_i^{r} are the cut-in, cut-out and rated wind speed (in m/s), respectively. a_3, a_2, a_1, and a_0 are the fitting coefficients.

Available photovoltaic power output is simply calculated by the solar irradiance-power conversion function:

$$P_{i,t}^{\mathrm{as}} = \eta_i^{\mathrm{s}} S_i I_{i,t} \quad \forall i,t, \tag{44}$$

where η_i^{s} is the photoelectric transformation efficiency of the photovoltaic array. S_i is the surface area of the photovoltaic array (in m^2). $I_{i,t}$ is the solar irradiance (in kW/m^2).

In this way, the upper and lower bounds of EM and ASM prices, as well as available wind and photovoltaic power output are acquired.

3.3. RO model formulation

As previous mentioned, the upper and lower bounds of uncertain parameters are derived from their predicted confidence intervals with the considered confidence level. Here, we set the confidence level equal to 100% to guarantee all feasible realization of uncertain parameters.

The objective function (1) includes all of uncertain factors of market prices. A general market price set J^{m} is presented to embrace all of these market price parameters as follows:

$$J^{\mathrm{m}} = \left\{\lambda_t^{\mathrm{e}} \in \left[\underline{\lambda}_t^{\mathrm{e}}, \overline{\lambda}_t^{\mathrm{e}}\right]; \lambda_t^{\mathrm{ru}} \in \left[\underline{\lambda}_t^{\mathrm{ru}}, \overline{\lambda}_t^{\mathrm{ru}}\right]; \lambda_t^{\mathrm{rd}} \in \left[\underline{\lambda}_t^{\mathrm{rd}}, \overline{\lambda}_t^{\mathrm{rd}}\right]; \lambda_t^{\mathrm{sr}} \in \left[\underline{\lambda}_t^{\mathrm{sr}}, \overline{\lambda}_t^{\mathrm{sr}}\right]; \lambda_t^{\mathrm{nsr}} \in \left[\underline{\lambda}_t^{\mathrm{nsr}}, \overline{\lambda}_t^{\mathrm{nsr}}\right]\right\}. \tag{45}$$

The market price uncertain parameters are set as their predicted confidence intervals. Note that the 100% confidence level leads to the most conservative solution. In order to adjust the level of robustness against price variations, we introduce a control parameter Γ^{m}, ranging from 0 to $|J^{\mathrm{m}}|$ ($|J^{\mathrm{m}}|$ denotes the cardinal number of set J^{m}). Because 120 uncertain parameters (24 price parameters each in energy, regulation up, regulation down, spinning reserve, and non-spinning reserve markets) are included in the uncertainty set J^{m}, $|J^{\mathrm{m}}|$ is equal to 120. As a consequence, the control parameter Γ^{m} can vary within the interval [0, 120]. If $\Gamma^{\mathrm{m}} = 0$, the influence of market price deviation from predicted mean values is ignored, and the optimization results are consistent with the deterministic model, while if $\Gamma^{\mathrm{m}} = 120$, the maximum influence of price deviation is considered, leading to the most conservative strategy.

Different from all of the market price parameters in the objective function, there is only one uncertain wind parameter (i.e., the available wind power output $P_{i,t}^{\mathrm{aw}}$) and one uncertain photovoltaic parameter (i.e., the available photovoltaic power output $P_{i,t}^{\mathrm{as}}$) in (15) and (16) for area i during hour h, respectively. Similarly, a wind uncertainty set $J_{i,t}^{\mathrm{w}}$ and a photovoltaic uncertainty set $J_{i,t}^{\mathrm{s}}$ are employed as follows:

$$J_{i,t}^{\mathrm{w}} = \left\{P_{i,t}^{\mathrm{aw}} \in \left[\underline{P}_{i,t}^{\mathrm{aw}}, \overline{P}_{i,t}^{\mathrm{aw}}\right]\right\} \quad \forall i,t. \tag{46}$$

$$J_{i,t}^{\mathrm{s}} = \left\{P_{i,t}^{\mathrm{as}} \in \left[\underline{P}_{i,t}^{\mathrm{as}}, \overline{P}_{i,t}^{\mathrm{as}}\right]\right\} \quad \forall i,t. \tag{47}$$

The control parameters $\Gamma_{i,t}^{\mathrm{w}}$ and $\Gamma_{i,t}^{\mathrm{s}}$, taking values within the interval $[0, \left|J_{i,t}^{\mathrm{w}}\right|]$ and $[0, \left|J_{i,t}^{\mathrm{s}}\right|]$, are used to control the robustness level against renewable power output deviation. Both $\left|J_{i,t}^{\mathrm{w}}\right|$ and $\left|J_{i,t}^{\mathrm{s}}\right|$ are equal to 1 because only one uncertain parameter lies on (15) or (16). In this regard, $\Gamma_{i,t}^{\mathrm{w}}$ and $\Gamma_{i,t}^{\mathrm{s}}$ are selected within the interval $[0, 1]$, and the level of robustness is enlarged as a higher value of $\Gamma_{i,t}^{\mathrm{w}}$ and $\Gamma_{i,t}^{\mathrm{s}}$.

Taking the above-mentioned uncertain parameters into account, the robust counterpart of the objective (1) and constraints (15) and (16) can be formulated as follows:

$$\begin{aligned} \min c(\Psi) + \max \sum_{t=1}^{T}\sum_{i=1}^{n_i}\Bigg\{\Bigg[&\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{e}} - \underline{\lambda}_t^{\mathrm{e}}\right)\left|P_{i,t}^{\mathrm{e}}\right| + \frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{ru}} - \underline{\lambda}_t^{\mathrm{ru}}\right)\left|R_{i,t}^{\mathrm{ru}}\right| \\ &+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{rd}} - \underline{\lambda}_t^{\mathrm{rd}}\right)\left|R_{i,t}^{\mathrm{rd}}\right| + \frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{sr}} - \underline{\lambda}_t^{\mathrm{sr}}\right)\left|R_{i,t}^{\mathrm{sr}}\right| \\ &+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{nsr}} - \underline{\lambda}_t^{\mathrm{nsr}}\right)\left|R_{i,t}^{\mathrm{nsr}}\right|\Bigg]\Bigg\} \end{aligned} \tag{48}$$

subject to

$$P_{i,t}^{\mathrm{w}} + \max\left[\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}} - \underline{P}_{i,t}^{\mathrm{aw}}\right)\left|p_{i,t}^{\mathrm{w}}\right|\right] \le \frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}} + \underline{P}_{i,t}^{\mathrm{aw}}\right) \quad \forall i,t \tag{49}$$

$$P_{i,t}^{\mathrm{s}}+\max\left[\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}-\underline{P}_{i,t}^{\mathrm{as}}\right)\left|p_{i,t}^{\mathrm{s}}\right|\right]\leq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}+\underline{P}_{i,t}^{\mathrm{as}}\right)\quad\forall i,t, \tag{50}$$

where $c(\Psi)$ denotes the objective function in the deterministic model. $p_{i,t}^{\mathrm{w}}$ and $p_{i,t}^{\mathrm{s}}$ are auxiliary variables to build a standard RO model and are forced to be equal to 1.

Incorporating control parameters into the model, (48)–(50) can be replaced by:

$$\begin{aligned}
\min c(\Psi)+max\Bigg\{&\sum_{t=1}^{T}\Bigg[\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{e}}-\underline{\lambda}_t^{\mathrm{e}}\right)\sum_{i=1}^{n_i}\left|P_{i,t}^{\mathrm{e}}\right|\upsilon_t^{\mathrm{e}}\\
&+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{ru}}-\underline{\lambda}_t^{\mathrm{ru}}\right)\sum_{i=1}^{n_i}\left|R_{i,t}^{\mathrm{ru}}\right|\upsilon_t^{\mathrm{ru}}+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{rd}}-\underline{\lambda}_t^{\mathrm{rd}}\right)\\
&\times\sum_{i=1}^{n_i}\left|R_{i,t}^{\mathrm{rd}}\right|\upsilon_t^{\mathrm{rd}}+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{sr}}-\underline{\lambda}_t^{\mathrm{sr}}\right)\sum_{i=1}^{n_i}\left|R_{i,t}^{\mathrm{sr}}\right|\upsilon_t^{\mathrm{sr}}\\
&+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{nsr}}-\underline{\lambda}_t^{\mathrm{nsr}}\right)\sum_{i=1}^{n_i}\left|R_{i,t}^{\mathrm{nsr}}\right|\upsilon_t^{\mathrm{nsr}}\Bigg]\\
&:\sum_{t=1}^{T}\left(\upsilon_t^{\mathrm{e}}+\upsilon_t^{\mathrm{ru}}+\upsilon_t^{\mathrm{rd}}+\upsilon_t^{\mathrm{sr}}+\upsilon_t^{\mathrm{nsr}}\right)\leq\Gamma^{\mathrm{m}},0\\
&\leq\upsilon_t^{\mathrm{e}},\upsilon_t^{\mathrm{ru}},\upsilon_t^{\mathrm{rd}},\upsilon_t^{\mathrm{sr}},\upsilon_t^{\mathrm{nsr}}\leq 1\Bigg\},
\end{aligned} \tag{51}$$

subject to

$$\begin{aligned}
&P_{i,t}^{\mathrm{w}}+\max\left[\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}}-\underline{P}_{i,t}^{\mathrm{aw}}\right)\left|p_{i,t}^{\mathrm{w}}\right|\upsilon_{i,t}^{\mathrm{w}}:\upsilon_{i,t}^{\mathrm{w}}\leq\Gamma_{i,t}^{\mathrm{w}},0\leq\upsilon_{i,t}^{\mathrm{w}}\leq 1\right]\\
&\leq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}}+\underline{P}_{i,t}^{\mathrm{aw}}\right)\quad\forall i,t
\end{aligned} \tag{52}$$

$$\begin{aligned}
&P_{i,t}^{\mathrm{w}}+\max\left[\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}-\underline{P}_{i,t}^{\mathrm{as}}\right)\left|p_{i,t}^{\mathrm{s}}\right|\upsilon_{i,t}^{\mathrm{s}}:\upsilon_{i,t}^{\mathrm{s}}\leq\Gamma_{i,t}^{\mathrm{s}},0\leq\upsilon_{i,t}^{\mathrm{s}}\leq 1\right]\\
&\leq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}+\underline{P}_{i,t}^{\mathrm{as}}\right)\quad\forall i,t.
\end{aligned} \tag{53}$$

The maximum problems formulated in (51)–(53) can be equally converted into their dual minimization problems taking the advantage of the strong duality theory as follows:

$$\begin{aligned}
\min\sum_{t=1}^{T}\sum_{i=1}^{n_i}\Bigg\{&-\Bigg[\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{e}}+\underline{\lambda}_t^{\mathrm{e}}\right)P_{i,t}^{\mathrm{e}}+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{ru}}+\underline{\lambda}_t^{\mathrm{ru}}\right)R_{i,t}^{\mathrm{ru}}+\frac{1}{2}\Big(\overline{\lambda}_t^{\mathrm{rd}}\\
&+\underline{\lambda}_t^{\mathrm{rd}}\Big)R_{i,t}^{\mathrm{rd}}+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{sr}}+\underline{\lambda}_t^{\mathrm{sr}}\right)R_{i,t}^{\mathrm{sr}}+\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{nsr}}+\underline{\lambda}_t^{\mathrm{nsr}}\right)R_{i,t}^{\mathrm{nsr}}\Bigg]+\Bigg(\lambda_i^{\mathrm{o}}\mu_{i,t}^{\mathrm{o}}\\
&+\frac{860\lambda_t^{\mathrm{ng}}}{HV^{\mathrm{ng}}}\times\frac{P_{i,t}^{\mathrm{gt}}}{\eta_i^{\mathrm{gt}}}+\lambda_i^{\mathrm{sb}}\mu_{i,t}^{\mathrm{sb}}+\lambda_i^{\mathrm{su}}\mu_{i,t}^{\mathrm{su}}+\lambda_i^{\mathrm{sd}}\mu_{i,t}^{\mathrm{sd}}\Bigg)+\Bigg(\frac{860\lambda_t^{\mathrm{ng}}}{HV^{\mathrm{ng}}}\\
&\times\frac{P_{i,t}^{\mathrm{gb}}}{\eta_i^{\mathrm{gb}}}\Bigg)+\Gamma^{\mathrm{m}}\upsilon^{\mathrm{m}}+\sum_{t=1}^{T}\left(q_t^{\mathrm{e}}+q_t^{\mathrm{ru}}+q_t^{\mathrm{rd}}+q_t^{\mathrm{sr}}+q_t^{\mathrm{nsr}}\right)\Bigg\},
\end{aligned} \tag{54}$$

subject to

$$\left\{\begin{aligned}
&\upsilon^{\mathrm{m}}+q_t^{\mathrm{e}}\geq\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{e}}-\underline{\lambda}_t^{\mathrm{e}}\right)y_t^{\mathrm{e}}\\
&\upsilon^{\mathrm{m}}+q_t^{\mathrm{ru}}\geq\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{ru}}-\underline{\lambda}_t^{\mathrm{ru}}\right)y_t^{\mathrm{ru}}\\
&\upsilon^{\mathrm{m}}+q_t^{\mathrm{rd}}\geq\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{rd}}-\underline{\lambda}_t^{\mathrm{rd}}\right)y_t^{\mathrm{rd}}\\
&\upsilon^{\mathrm{m}}+q_t^{\mathrm{sr}}\geq\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{sr}}-\underline{\lambda}_t^{\mathrm{sr}}\right)y_t^{\mathrm{sr}}\\
&\upsilon^{\mathrm{m}}+q_t^{\mathrm{nsr}}\geq\frac{1}{2}\left(\overline{\lambda}_t^{\mathrm{nsr}}-\underline{\lambda}_t^{\mathrm{nsr}}\right)y_t^{\mathrm{nsr}}\\
&-y_t^{\mathrm{e}}\leq\sum_{i}^{n_i}P_{i,t}^{\mathrm{e}}\leq y_t^{\mathrm{e}}\\
&-y_t^{\mathrm{ru}}\leq\sum_{i}^{n_i}R_{i,t}^{\mathrm{ru}}\leq y_t^{\mathrm{ru}}\\
&-y_t^{\mathrm{rd}}\leq\sum_{i}^{n_i}R_{i,t}^{\mathrm{rd}}\leq y_t^{\mathrm{rd}}\\
&-y_t^{\mathrm{sr}}\leq\sum_{i}^{n_i}R_{i,t}^{\mathrm{sr}}\leq y_t^{\mathrm{sr}}\\
&-y_t^{\mathrm{nsr}}\leq\sum_{i}^{n_i}R_{i,t}^{\mathrm{nsr}}\leq y_t^{\mathrm{nsr}}\\
&\upsilon^{\mathrm{m}},q_t^{\mathrm{e}},q_t^{\mathrm{ru}},q_t^{\mathrm{rd}},q_t^{\mathrm{sr}},q_t^{\mathrm{nsr}},\\
&y_t^{\mathrm{e}},y_t^{\mathrm{ru}},y_t^{\mathrm{rd}},y_t^{\mathrm{sr}},y_t^{\mathrm{nsr}}\geq 0
\end{aligned}\right.\qquad\forall t \tag{55}$$

$$\left\{\begin{aligned}
&P_{i,t}^{\mathrm{w}}+\Gamma_{i,t}^{\mathrm{w}}\upsilon_{i,t}^{\mathrm{w}}+q_{i,t}^{\mathrm{w}}\leq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}}+\underline{P}_{i,t}^{\mathrm{aw}}\right)\\
&P_{i,t}^{\mathrm{s}}+\Gamma_{i,t}^{\mathrm{s}}\upsilon_{i,t}^{\mathrm{s}}+q_{i,t}^{\mathrm{s}}\leq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}+\underline{P}_{i,t}^{\mathrm{as}}\right)\\
&\upsilon_{i,t}^{\mathrm{w}}+q_{i,t}^{\mathrm{w}}\geq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{aw}}-\underline{P}_{i,t}^{\mathrm{aw}}\right)y_{i,t}^{\mathrm{w}}\\
&\upsilon_{i,t}^{\mathrm{s}}+q_{i,t}^{\mathrm{s}}\geq\frac{1}{2}\left(\overline{P}_{i,t}^{\mathrm{as}}-\underline{P}_{i,t}^{\mathrm{as}}\right)y_{i,t}^{\mathrm{s}}\\
&y_{i,t}^{\mathrm{w}},y_{i,t}^{\mathrm{s}}\geq 1\\
&\upsilon_{i,t}^{\mathrm{w}},\upsilon_{i,t}^{\mathrm{s}},q_{i,t}^{\mathrm{w}},q_{i,t}^{\mathrm{s}}\geq 0
\end{aligned}\right.\qquad\forall i,t, \tag{56}$$

where υ^{m}, q_t^{e}, q_t^{ru}, q_t^{rd}, q_t^{sr}, and q_t^{nsr} are dual variables related to the objective function (51). $\upsilon_{i,t}^{\mathrm{w}}$, $\upsilon_{i,t}^{\mathrm{s}}$, $q_{i,t}^{\mathrm{w}}$, and $q_{i,t}^{\mathrm{s}}$ are dual variables related to constraints (52) and (53). y_t^{e}, y_t^{ru}, y_t^{rd}, y_t^{sr}, y_t^{nsr}, $y_{i,t}^{\mathrm{w}}$, and $y_{i,t}^{\mathrm{s}}$ are auxiliary variables used to linearize the absolute function $|\cdot|$.

Consequently, the deterministic MILP model (1)–(34) can be transformed into a robust MILP model (2)–(14), (17)–(34), and (54)–(56).

The procedure for the proposed RO model of the ICES is presented in Fig. 3. We solve this model in two steps. In the first step, we obtain confidence intervals of uncertain parameters by the GP and the power conversion function, performed on MATLAB. In the second step, we solve the robust MILP model by the CPLEX solver on GAMS.

4. Case study

4.1. Data description

The proposed method is tested on a real regional multi-energy system, which achieves CCHP for industrial, commercial, and

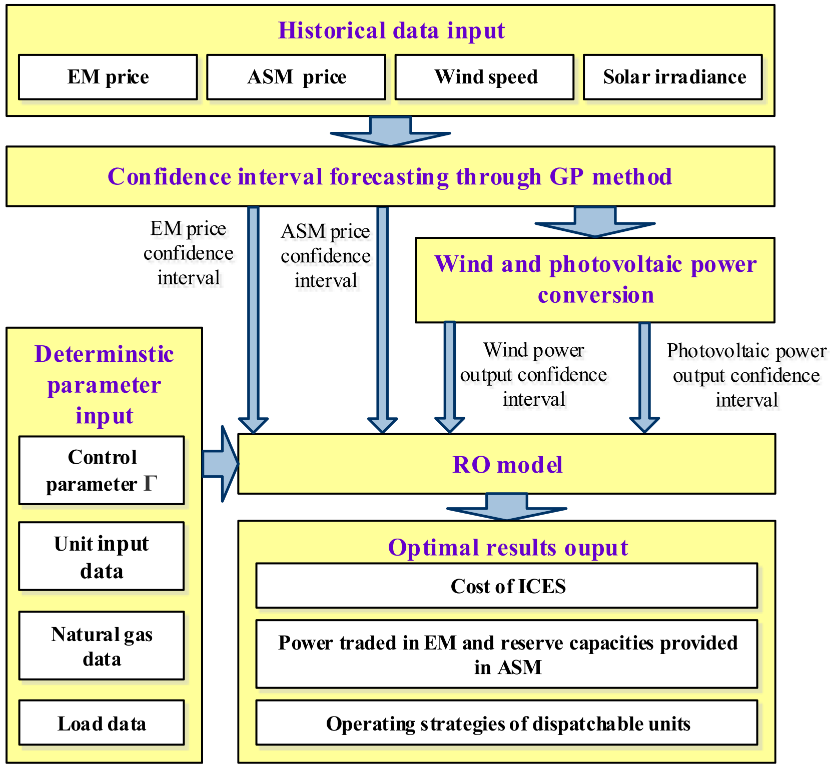


Fig. 3. Procedure for the proposed RO model of an ICES using the GP.

Table 1
Input parameters for units of ICES.

$P_i^{gt,\max}$, $P_i^{gt,\min}$ (kW)	r_i^{u}, r_i^{d} (kW/h)	λ_i^{o}, λ_i^{su}, λ_i^{sd}, λ_i^{sb} (\$)	$T_i^{\mathrm{u,min}}T_i^{\mathrm{d,min}}$ (h)	η_i^{gt}, η_i^{gtl}
600, 120	1800, 1800	3.4, 0.09, 0.08, 0.03	2, 2	0.35, 0.1
$H_i^{\mathrm{gb,max}}$, $H_i^{\mathrm{gb,min}}$ (kW)	η_i^{gb}, η_i^{hrs}	$P_i^{\mathrm{ec,max}}$, $P_i^{\mathrm{ec,min}}$ (kW)	$H_i^{\mathrm{ac,max}}$, $H_i^{\mathrm{ac,min}}$ (kW)	COP_i^{ec}, COP_i^{ac}
800, 0	0.75, 0.75	420, 0	500, 0	4, 1.2

$P_i^{\mathrm{ce,max}}$, $P_i^{\mathrm{de,max}}$ (kW)	$S_{i,t}^{\mathrm{ees,max}}$, $S_{i,t}^{\mathrm{ees,min}}$ (kWh)	η_i^{ce}, η_i^{de}	$H_i^{\mathrm{ct,max}}$, $H_i^{\mathrm{dt,max}}$ (kW)
200, 200	400, 0	0.9, 0.9	230, 230
$S_{i,t}^{\mathrm{tes,max}}$, $S_{i,t}^{\mathrm{tes,min}}$ (kWh)	η_i^{ce}, η_i^{te}	$H_{ij}^{\mathrm{tf,max}}$, $Q_{ij}^{\mathrm{cf,max}}$ (kW)	η^{tf}, η^{cf} (km^{-1})
400, 0	0.98, 0.98	600, 600	0.02, 0.01

residential areas [15]. The required technical profiles for units are listed in Table 1 [15,18]. The distance between any pair of areas is assumed to be 3 km. The natural gas price is fixed as 0.187 \$/m^3 for summer and 0.156 \$/m^3 for winter in 2016 from the Ontario Energy Board [43]. The heating value of natural gas is 10,852 kcal/m^3 [44].

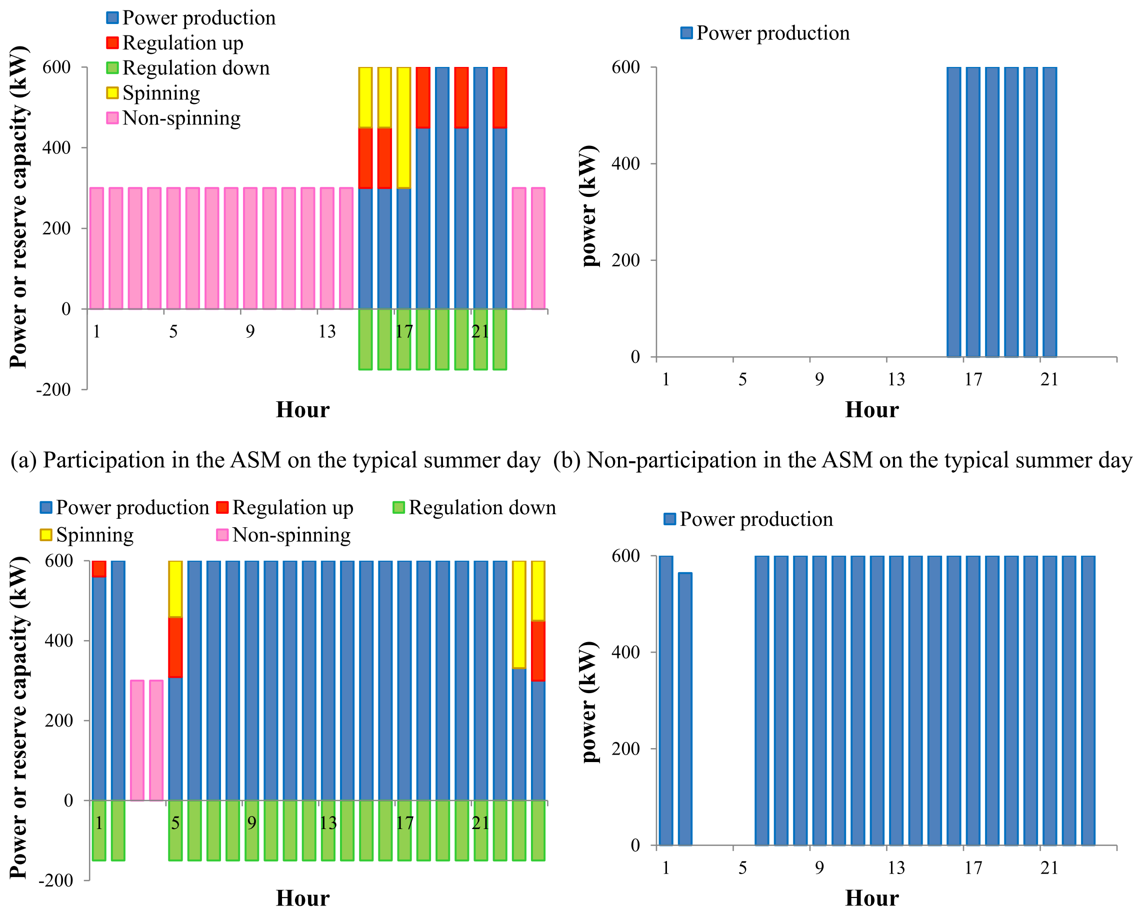
For comprehensiveness, the proposed model is formulated on both summer and winter days. A typical summer day, August 26, 2016, and a typical winter day, January 26, 2016, are chosen as the scheduling days. Historical data of EM and ASM prices in the California day-ahead market [45], as well as wind speed and solar irradiance, from August 1–25, 2016 and from January 1–25, 2016 are used to forecast the uncertain parameters on August 26, 2016 and January 26, 2016, respectively. The results of predicted confidence intervals and mean values of uncertain parameters are presented in Appendix A, which also presents electrical, heating, and cooling loads on the typical summer and winter days [15].

The wind power conversion curve is derived from a 1.5-MW wind turbine (Goldwind GW 77/1500 [46]). Its cut-in, cut-out, and rated wind speeds are 3, 22, and 11.5 m/s, respectively, and its fitting coefficients are $a_0 = 0.498$, $a_1 = -0.309$, $a_2 = 0.0594$, and $a_3 = -0.00249$. The photoelectric transformation efficiency is set to 15.7% and the surface area of the photovoltaic arrays in each area is assumed to be 2000 m [15].

4.2. Impact of ICES participation in the ASM

In this subsection, we examine the impact of the ICES participation in the ASM using the deterministic model. In order to analyze this impact on operational strategies of the gas turbine, optimization results of the gas turbine for the commercial area are presented in Fig. 4.

Although optimization results exhibit significant differences on the typical summer and winter days (e.g., the gas turbine exhibits higher amounts of production on the winter day owing to a higher heating load), a similar tendency in the comparison of whether or not the ICES participating in the ASM could be observed. For example, during some low-EM-price hours (hours 15 and 22 on the summer day and hours 5 and 24 on the winter day), the total profit in the EM and ASM is higher than the gas turbine cost. Therefore, the gas turbine is on and operates to provide more regulation up, regulation down, and spinning reserve capacities when the ICES participates in the ASM. However, because the profit in the EM is



(a) Participation in the ASM on the typical summer day (b) Non-participation in the ASM on the typical summer day

(c) Participation in the ASM on the typical winter day (d) Non-participation the ASM on the typical winter day

Fig. 4. Optimal results of gas turbine operation strategies for commercial area on the typical summer and winter days when ICES participation and non-participation in the ASM.

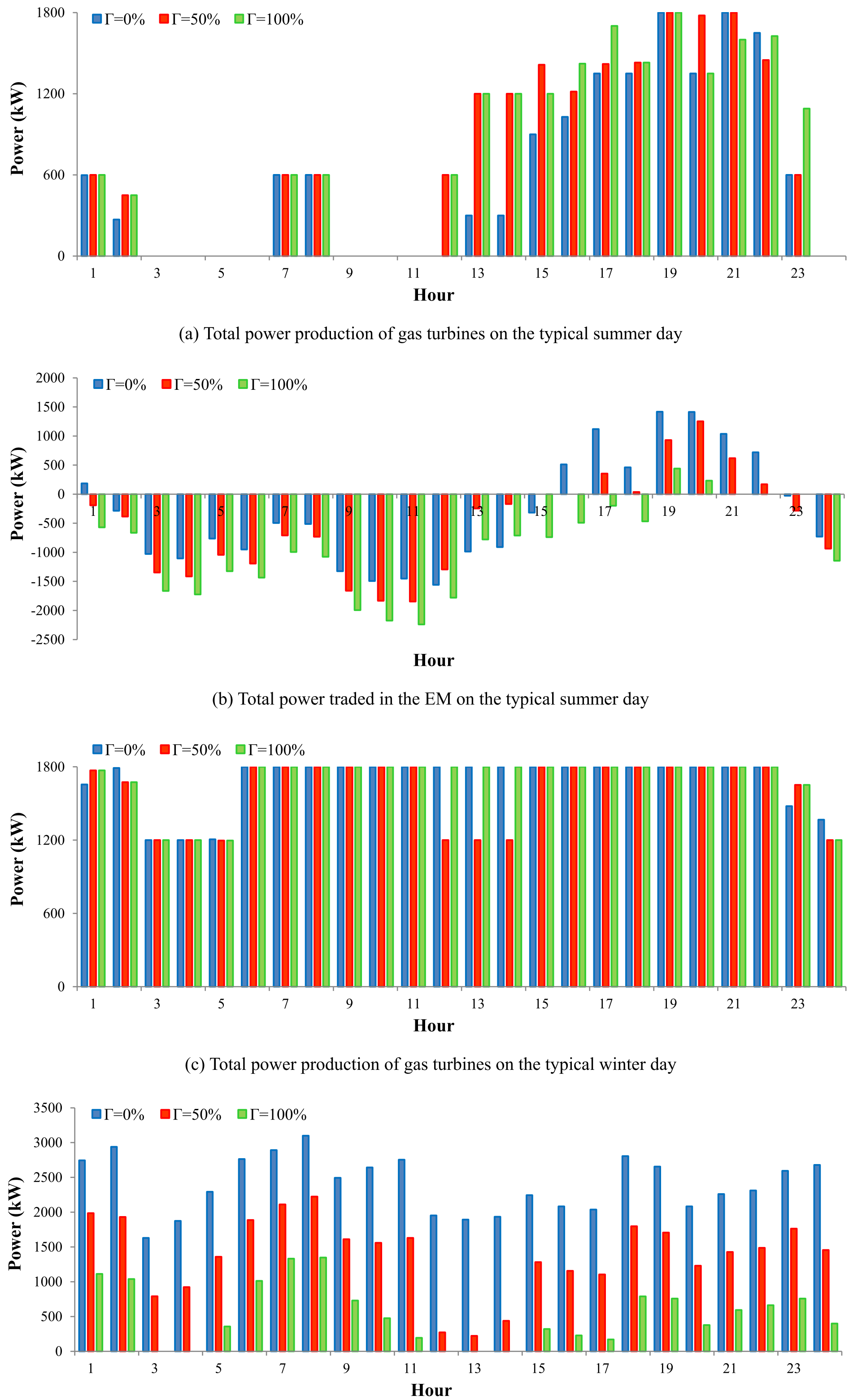
Table 2
Comparison of total cost (in $) for different cases on summer and winter days.

Day	No ASM	Spinning reserve market	Regulation reserve market	ASM including four types of reserves
August 26	813.41	802.36	786.79	783.76
August 27	1178.37	1171.71	1155.84	1151.60
August 28	985.97	980.73	967.95	965.80
August 29	1148.98	1142.79	1126.13	1124.52
August 30	831.55	825.03	807.41	806.29
August 31	1109.54	1107.21	1086.52	1084.98
January 26	196.45	195.79	159.51	159.28
January 27	919.59	919.23	881.07	881.07
January 28	1117.67	1117.29	1073.89	1073.89
January 29	1135.87	1134.94	1074.27	1074.27
January 30	817.31	816.58	762.97	762.53
January 31	446.54	445.64	357.65	357.44
average	891.77	888.27	853.33	852.12

lower than the gas turbine cost, the gas turbine is still off in the case of non-participation in the ASM. During some high-EM-price hours (hours 16–18 and 20 on the summer day and hour 23 on the winter day), the EM profit is higher than the gas turbine cost, but it is still lower than the ASM profit plus the gas turbine cost. Consequently, when the ICES participates in the ASM, the gas turbine operates at a lower level to provide more regulation up and spinning reserve capacities. In contrast, when the ICES does not participate in the ASM, it operates at the maximum level to sell more power in the EM. It is noteworthy that the regulation up reserve outperforms the spinning reserve during most hours because of a higher price reward for the regulation up reserve. Moreover, the gas turbine offers the regulation down reserve during all on hours and the non-spinning reserve during all off hours to obtain more ASM revenue.

To further evaluate the influence of ICES participation in the ASM, an "out of sample" analysis is performed based on the predicted data from August 26–31 (6 summer days) and January 26–31 (6 winter days).

The total cost of participation in the ASM is lowest on each summer and winter day (Table 2). Specifically, the average total cost



(a) Total power production of gas turbines on the typical summer day

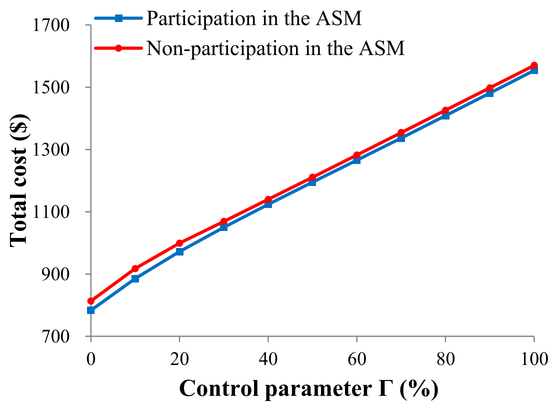
(b) Total power traded in the EM on the typical summer day

(c) Total power production of gas turbines on the typical winter day

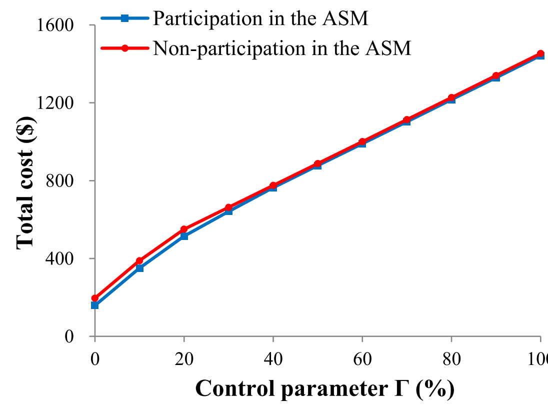
(d) Total power traded in the EM on the typical winter day

Fig. 5. Effect of control parameter Γ on total power production of gas turbines and total power traded in the EM.

of participation in the ASM is 1.42%, 4.07%, and 4.45% lower than participation in the regulation reserve market, participation in the spinning reserve market, and non-participation in the ASM, respectively. This is because of more flexibility obtained to provide profitable reserve capacities when the ICES participates in the ASM.



(a) Total cost on the typical summer day



(b) Total cost on the typical winter day

Fig. 6. Effect of control parameter Γ on the total cost when participation and not participation in the ASM.

4.3. Impact of the RO strategy

The conservation level could be adjusted by the values of the control parameters Γ^{m}, $\Gamma^{w}_{i,t}$, and $\Gamma^{s}_{i,t}$. For simplicity, all of the control parameters are expressed as a percentage of their maximum value and are represented by a same control parameter Γ, i.e., $\Gamma = \Gamma^{m} = \Gamma^{w}_{i,t} = \Gamma^{s}_{i,t}$.

Fig. 5 presents the total gas turbine power production and the total power traded in the EM for all of the areas when different values of Γ are chosen. The total power sold in the EM decreases with a higher value of Γ on both the typical summer and winter days. Indeed, a higher value of Γ results in a more conservative strategy. This strategy encourages the ICES to sell less power to avoid possible economic losses caused by the uncertainty of renewable power output and market prices. In comparison, the impact of Γ on the gas turbine generation differs between summer and winter days. This is because gas turbines respond differently to the deviation of renewable power output and market prices. Specifically, the gas turbine generation increases when more influence of renewable power output deviation is considered. Nevertheless, it decreases when more influence of market price deviation is considered. This is because the decrease of the gas turbine generation results in a reduced amount of power sold in the EM, which can avoid the risk of selling power in a low EM price. We note that the ICES purchases power during most of the hours in the typical summer day, whereas it sells power during all of the hours in the typical winter day (Fig. 5(b) and (d)). Therefore, when the value of Γ increases from 0% to 50%, the total generation of gas turbines increases against renewable power output deviation during hours 2, 12–18, and 20 on the typical summer day, whereas it decreases against market price deviation during hours 12–14 on the typical winter day. Moreover, the total gas turbine generation on the winter day is higher at Γ = 100% than at Γ = 50%, owing to the fact that renewable power output suffers from a higher level of uncertainty.

Operation strategies in different cases of Γ are summarized in Table 3. Note that the results are the sum of power or reserve capacities for all of the areas on the scheduling day. The results presented in this table are in accord with the analysis of Fig. 5, and the reserve capacities correlate with the power production of gas turbines. For example, as a result of the gas turbine power production increases on the summer day, the regulation up and spinning reserve capacities decrease, and the regulation down capacity increases. Moreover, the ICES provides no non-spinning reserve capacity in a more conservative strategy (when Γ = 50% and Γ = 100%) to avoid the condition in which the profit in the non-spinning reserve market is lower than the standby cost.

The level of robustness increases with Γ at the expense of profitability losses. The total cost associated with the value variability of Γ is illustrated in Fig. 6, which corroborates that more total cost corresponds to a higher value of Γ, irrespective of the scheduling day and the situation of participation or non-participation in the ASM. Specifically, when the value of Γ increases from 0 to 100%, the total cost increases by $769.96 (from $783.76 to $1553.73) on the summer day and $1283.65 (from $159.28 to $1442.93) on the winter day with the ICES participation in the ASM. However, if the ICES participation in the ASM is not considered, the incremental cost is smaller (an increase of $756.80 on the summer day and of $1257.15 on the winter day). This is because with participation in the ASM, economic losses become more conspicuous to cope with more uncertainty, namely, the uncertainty of ASM prices. Additionally, the gap of total cost between participation and non-participation in the ASM decreases with a value increment of Γ as a result of lower potential income in the ASM to achieve robustness.

Table 3
Results comparison of operation strategies in different cases on the typical summer and winter days.

Day	Value of Γ	Power traded in the EM (kW)	Power production of gas turbines (kW)	Regulation up capacity (kW)	Regulation down capacity (kW)	Spinning reserve capacity (kW)	Non-spinning reserve capacity (kW)
Summer	Γ = 0%	−7081	14,497	2402	4650	1671	12,300
	Γ = 50%	−11,926	18,158	1658	5100	584	0
	Γ = 100%	−21,510	18,470	1501	5100	429	0
Winter	Γ = 0%	57,668	40,497	888	10,500	616	600
	Γ = 50%	33,364	38,693	0	9750	303	0
	Γ = 100%	12,663	40,493	0	10,200	303	0

5. Conclusions

This paper proposes a day-ahead scheduling strategy for the CCHP-based ICES that participates in the joint EM and ASM, with the RO approach to handle the uncertainty of market prices and renewable power output. In this model, we consider regulation up, regulation down, spinning, and non-spinning reserves in the ASM. The proposed strategy is established as the robust MILP model and simulated for the real regional multi-energy system, which demonstrates the effectiveness and applicability of the proposed model.

Considering the case study's numerical results, major conclusions of this paper are threefold:

1) Participation in the ASM leads to a reduction of total cost on both summer and winter days since gas turbines operate at a lower level with longer hours to provide more ancillary service capacities.
2) The trade-off between robustness and profitability can be adjusted by the control parameter of the RO strategy, serving as a flexibility approach for the ICES to address the challenge posed by uncertainty.
3) ICES participation in the ASM suffers more economic losses for robustness, but the total cost of the ICES is still lower than the case of non-participation in the ASM.

Owing to that the predicted accuracy affects the optimal results, in practical application, the performance of the RO model can be improved by an extension of standard GP and a continuous update of historical data. A multi-stage procedure in the day-ahead and real-time markets represents the direction of our future research.

Acknowledgment

This work is supported by the National Science Foundation of China (No. 51507052) and the Science and Technology Project of the State Grid Corporation of China (No. J2017129).

Appendix A

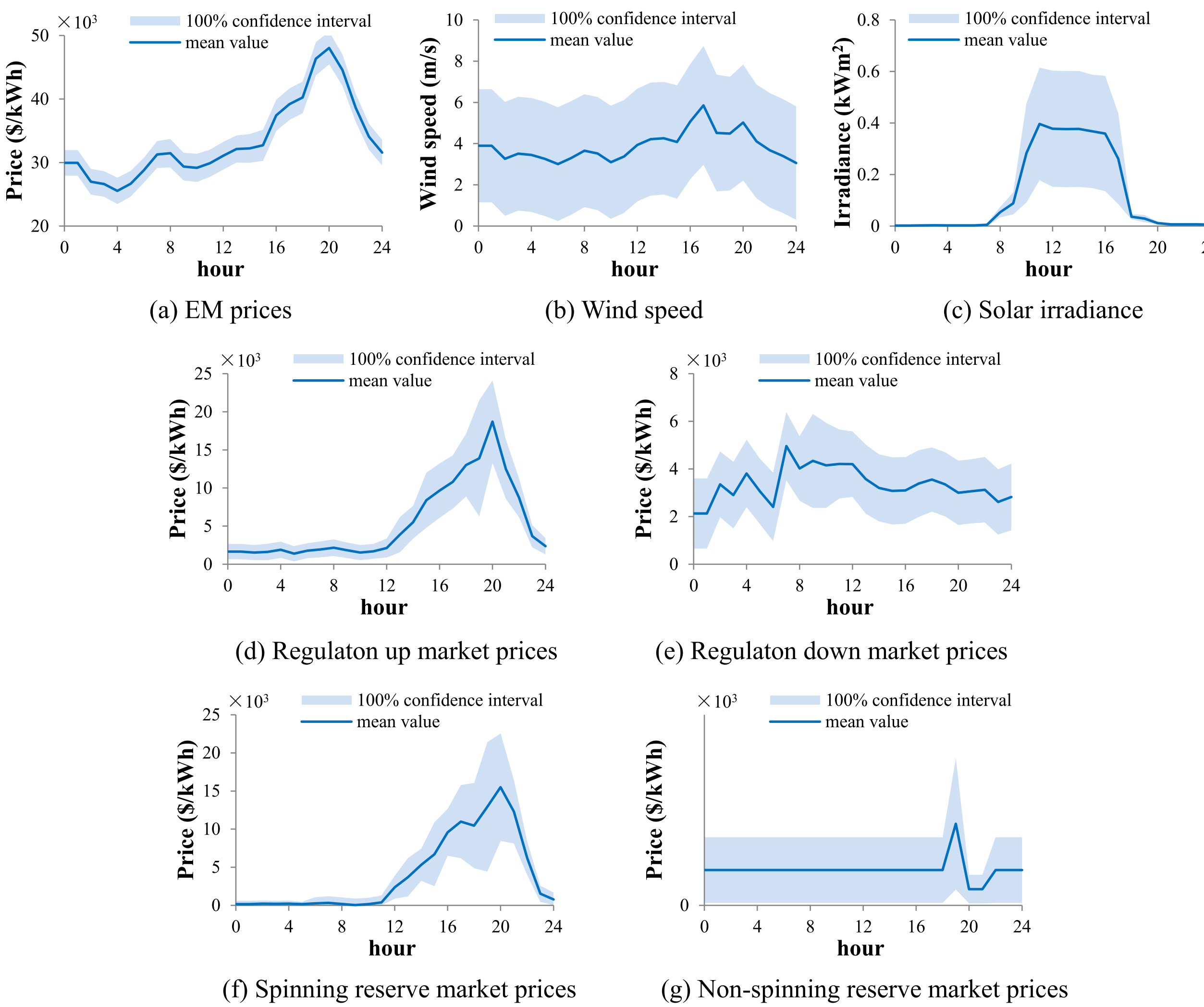


Fig. A1. Predicted confidence intervals and mean values of EM prices, ASM prices, wind speed and solar irradiance on the typical summer day.

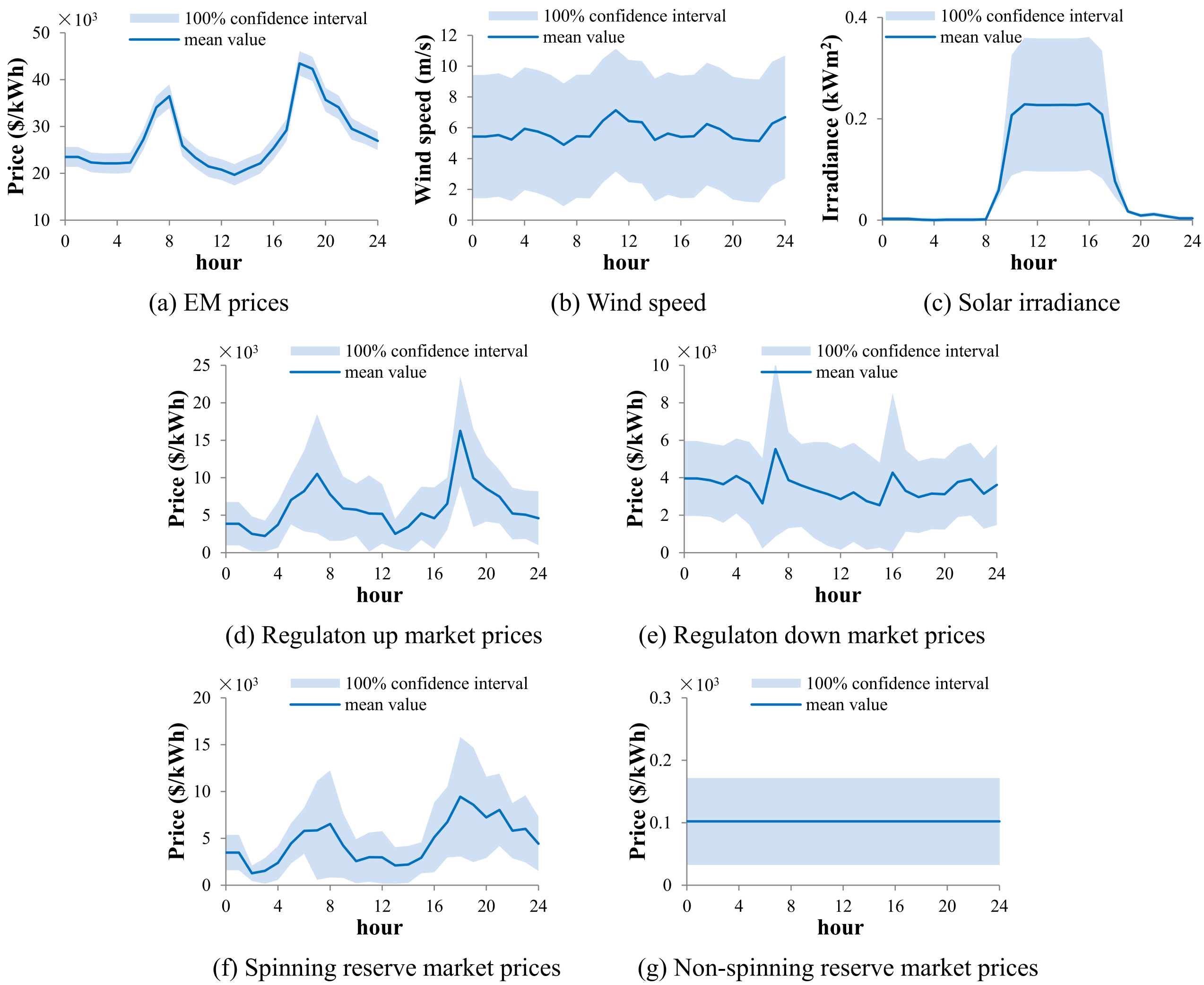


Fig. A2. Predicted confidence intervals and mean values of EM prices, ASM prices, wind speed and solar irradiance on the typical winter day.

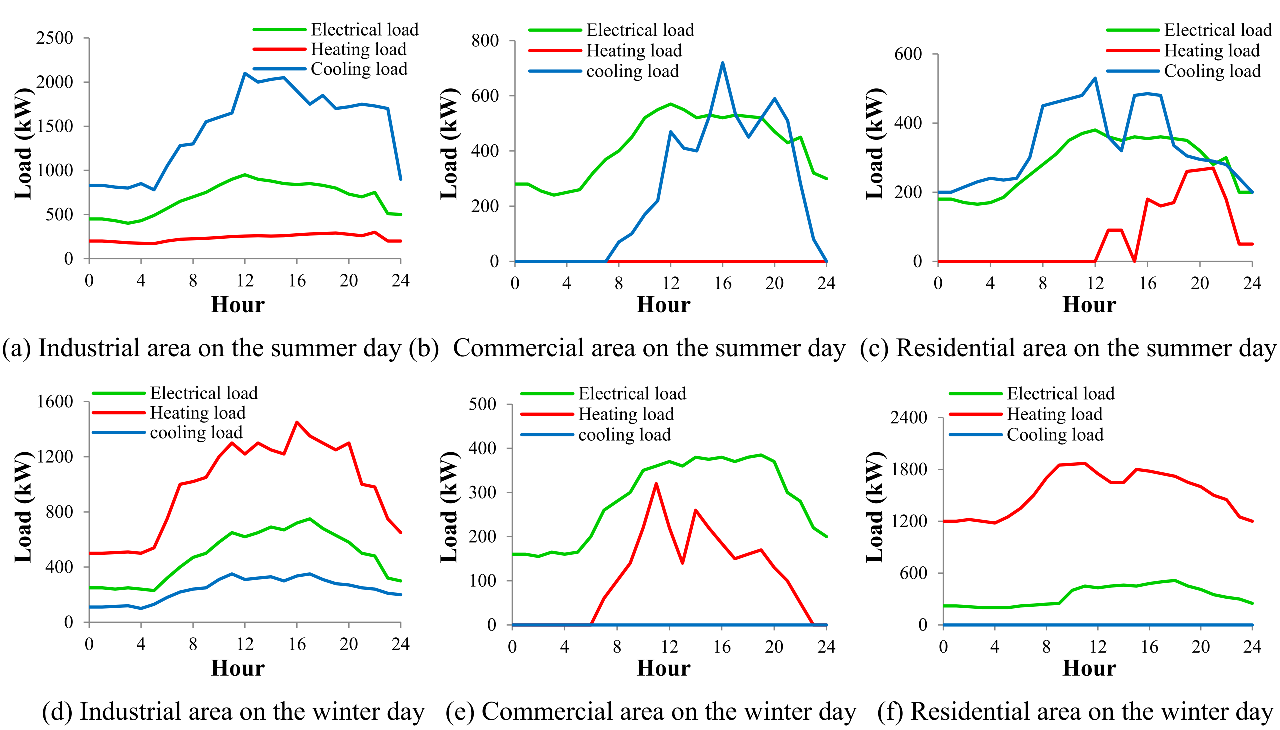


Fig. A3. Electrical, heating and cooling loads for industrial, commercial and residential areas on the typical summer and winter days.

References

[1] Moghaddam AA, Seifi A, Niknam T, Pahlavani MRA. Multi-objective operation management of a renewable mg (micro-grid) with back-up micro-turbine/fuel cell/battery hybrid power source. Energy 2011;36(11):6490–507.

[2] Robu V, Chalkiadakis G, Kota R, Rogers A, Jennings NR. Rewarding cooperative virtual power plant formation using scoring rules. Energy 2016;117:19–28.

[3] Bagdanavicius A, Jenkins N, Hammond GP. Assessment of community energy supply systems using energy, exergy and exergoeconomic analysis. Energy 2012;45:247–55.

[4] Ersoz I, Colak U. Combined cooling, heat and power planning under uncertainty. Energy 2016;24(1):1016–25.

[5] Eid C, Bollinger LA, Koirala B, Scholten D, Facchinetti E, Lilliestam J, et al. Market integration of local energy systems: is local energy management compatible with European regulation for retail competition? Energy 2016;114:913–22.

[6] Lund H, Andersen AN, Østergaard PA, Mathiesen BV, Connolly D. From electricity smart grids to smart energy systems – a market operation based approach and understanding. Energy 2012;42:96–102.

[7] Koirala BP, Koliou E, Friege J, Hakvoort RA, Herder PM. Energetic communities for community energy: a review of key issues and trends shaping integrated community energy systems. Renew Sustain Energy Rev 2016;56:722–44.

[8] Mendes G, Loakimidis C, Ferraro P. On the planning and analysis of integrated community energy systems: a review and survey of available tools. Renew Sustain Energy Rev 2011;15:4836–54.

[9] Bakken BH, Skjelbred HI, Wolfgang O. eTransport: investment planning in energy supply systems with multiple energy carriers. Energy 2007;32:1676–89.

[10] Li G, Zhang R, Jiang T, Chen H, Bai L, Cui H, et al. Optimal dispatch strategy for integrated energy systems with CCHP and wind power. Appl Energy 2017;192:408–19.

[11] Xu X, Jin X, Jia H, Yu X, Li K. Hierarchical management for integrated community energy systems. Appl Energy 2015;160:231–43.

[12] Motevasel M, Seifi AR, Niknam T. Multi-objective energy management of CHP (combined heat and power)-based micro-grid. Energy 2013;51:123–36.

[13] Zamani AG, Zakariazadeh A, Jadid S, Kazemi A. Stochastic operational scheduling of distributed energy resources in a large scale virtual power plant. Electr Power Energy Syst 2016;82:608–20.

[14] Wu R, Mavromatidis G, Orehounig K, Carmeliet J. Multiobjective optimisation of energy systems and building envelope retrofit in a residential community. Appl Energy 2017;190:634–49.

[15] Yang H, Xiong T, Qiu J, Qiu D, Dong ZY. Optimal operation of DES/CCHP based regional multi-energy prosumer with demand response. Appl Energy 2016;167:353–65.

[16] González P, Villar J, Díaz CA, Campos FA. Joint energy and reserve markets: current implementations and modeling trends. Elec Power Syst Res 2014;109:101–11.

[17] Haghighat H, Seifi H, Kian AR. Gaming analysis in joint energy and spinning reserve markets. IEEE Trans Power Syst 2007;22(4):2074–85.

[18] Shi L, Luo Y, Tu GY. Bidding strategy of microgrid with consideration of uncertainty for participating in power market. Elec Power Energy Syst 2014;59:1–13.

[19] Zamani AG, Zakariazadeh A, Jadid S. Day-ahead resource scheduling of a renewable energy based virtual power plant. Appl Energy 2016;169:324–40.

[20] Nazari ME, Ardehali MM. Profit-based unit commitment of integrated CHP-thermal-heat only units in energy and spinning reserve markets with considerations for environmental CO_2 emission cost and valve-point effects. Energy 2017;133:621–35.

[21] Pousinho HMI, Silva H, Mendes VMF, Collares-Pereira M, Cabrita CP. Self-scheduling for energy and spinning reserve of wind/CSP plants by a MILP approach. Energy 2014;78:524–34.

[22] Shayegan-Rad A, Badri A, Zangeneh A. Day-ahead scheduling of virtual power plant in joint energy and regulation reserve markets under uncertainties. Energy 2017;121:114–25.

[23] Al-Swaiti MS, Al-Awami AT, Khalid MW. Co-optimized trading of wind-thermal-pumped storage system in energy and regulation markets. Energy 2017;138:991–1005.

[24] He G, Chen Q, Kang C, Xia Q, Poolla K. Cooperation of wind power and battery storage to provide frequency regulation in power markets. IEEE Trans Power Syst 2017;32(5):3559–68.

[25] Bertsimas D, Sim M. The price of robustness. Oper Res 2004;52:35–53.

[26] Bertsimas D, Sim M. Robust discrete optimization and network flows. Math Program 2003;98:49–71.

[27] Bertsimas D, Brown DB, Caramanis C. Theory and applications of robust optimization. SIAM Rev 2011;53(3):464–501.

[28] Xiong P, Jirutitijaroen P, Singh C. A distributionally robust optimization model for unit commitment considering uncertain wind power production. IEEE Trans Power Syst 2017;31(2):39–49.

[29] Martinez-Mares A, Fuerte-Esquivel CR. A robust optimization approach for the interdependency analysis of integrated energy systems considering wind power uncertainty. IEEE Trans Power Syst 2013;28(4):3964–76.

[30] Baringo L, Conejo AJ. Offering strategy via robust optimization. IEEE Trans Power Syst 2011;26(3):1418–25.

[31] Shabanzadeh M, Sheikh-El-Eslami M, Haghifam M. The design of a risk-hedging tool for virtual power plants via robust optimization approach. Appl Energy 2015;155:766–77.

[32] Wang R, Wang P, Xiao G. A robust optimization approach for energy generation scheduling in microgrids. Energy Convers Manag 2015;106:597–607.

[33] Rahimiyan M, Baringo L. Strategic bidding for a virtual power plant in the day-ahead and real-time markets: a price-taker robust optimization approach. IEEE Trans Power Syst 2016;31(4):2676–87.

[34] Liu M, Shi Y, Fang F. Combined cooling, heating and power systems: a survey. Renew Sustain Energy Rev 2014;35:1–22.

[35] Cho H, Smith AD, Mago P. Combined cooling, heating and power systems: a review of performance improvement and optimization. Appl Energy 2014;136:168–85.

[36] Lund H, Möller B, Mathiesen BV, Dyrelund A. The role of district heating in future renewable energy systems. Energy 2010;35:1381–90.

[37] Bayindir R, Colak I, Fulli G, Demirtas K. Smart grid technologies and applications. Renew Sustain Energy Rev 2016;66:499–516.

[38] California ISO: market processes and products. <http://www.caiso.com/market/Pages/MarketProcesses.aspx> [accessed 01.03.2017].

[39] California ISO, Pay for Performance Regulation Draft Final Proposal. <http://www.caiso.com/Documents/DraftFinalProposal-PayforPerformanceRegulation-YearOneDesign.pdf>. [accessed 20.05.17].

[40] Rasmussen CE, Williams CKI. Gaussian processes for machine learning. Cambridge, MA: MIT Press; 2006.

[41] Chen N, Qian Z, Nabney IT, Meng X. Wind power forecasts using Gaussian processes and numerical weather prediction. IEEE Trans Power Syst 2014;29(2):656–65.

[42] Kou P, Liang D, Gao L, Lou J. Probabilistic electricity price forecasting with variational heteroscedastic Gaussian process and active learning. Energy Convers Manag 2015;89:298–308.

[43] Natural Gas Rates Historical. <http://www.ontarioenergyboard.ca/oeb/Consumers/Natural%20Gas/Natural%20Gas%20Rates> [accessed 20.03.17].

[44] Natural Gas: A Primer <http://www.nrcan.gc.ca/energy/natural-gas/5641> [accessed 20.03.17].

[45] Electric Power Markets: California (CAISO). <https://www.ferc.gov/market-oversight/mkt-electric/california.asp> [accessed 20.03.17].

[46] Goldwind GW 77/1500. < https://en.wind-turbine-models.com/turbines/106-goldwind-gw-77-1500>. [accessed 02.04.17].

Nomenclature

Indices

t: Index of time periods in 1 h, running from 1 to T
i, j: Index of areas, running from 1 to n_i

Parameters and constants

$P_i^{gt,max}/P_i^{gt,min}$: Maximum/minimum electrical power production of gas turbine (kW)
r_i^u/r_i^d: Ramp-up/ramp-down limit of gas turbine (kW/h)
$\lambda_i^{f/su/sd/sb}$: Fix/startup/shutdown/standby cost of gas turbine ($)
$T_i^{u,nin}/T_i^{d,min}$: Minimum up/down time of gas turbine (h)
η_i^{gt}: Efficiency for electrical power production of gas turbine
η_i^{gtl}: Energy losses of gas turbine
$H_i^{gb,max}/H_i^{gb,min}$: Maximum/minimum thermal power production of gas boiler (kW)
η_i^{gb}/η_i^{hrs}: Efficiency for thermal power production of gas boiler/heat recover system
$P_i^{ec,max}/P_i^{ec,min}$: Maximum/minimum electrical power input of electric chiller (kW)
$H_i^{ac,max}/H_i^{ac,min}$: Maximum/minimum thermal power input of absorption chiller (kW)
COP_i^{ec}/COP_i^{ac}: Coefficient for the performance of electric/absorption chiller
$P_i^{ce,max}/P_i^{de,max}$: Maximum charge/discharge electrical power of electrical energy storage (kW)
$S_{i,t}^{ees,max}/S_{i,t}^{ees,min}$: Maximum/minimum allowed electrical energy stored in electrical energy storage (kWh)
η_i^{ce}/η_i^{de}: Charge/discharge efficiency for electrical energy storage
$H_i^{ct,max}/H_i^{dt,max}$: Maximum charge/discharge thermal power of thermal energy storage (kW)
$S_{i,t}^{tes,max}/S_{i,t}^{tes,min}$: Maximum/minimum allowed thermal energy stored in thermal energy storage (kWh)
η_i^{ct}/η_i^{dt}: Charge/discharge efficiency for thermal energy storage
$H_{i,j}^{tf,max}/Q_{i,j}^{cf,max}$: Maximum thermal/cooling power flow exchanged between area i and area j (kW)
$d_{i,j}$: Distance between area i and area j (km)
η^{tf}/η^{cf}: Losses along thermal and cooling pipelines per length unit (km^{-1})
$P_{i,t}^{el}/H_{i,t}^{tl}/Q_{i,t}^{cl}$: Electrical/heating/cooling loads (kW)

$\lambda_t^{em/rum/rdm/srm/nsrm}$: Market price in energy/regulation up/regulation down/spinning reserve/non-spinning reserve market ($/kWh)
λ_t^{ng}: Natural gas price ($/m^3)
HV^{ng}: Heating value of natural gas (kcal/m^3)
$P_{i,t}^{aw}$: Available wind power output (kW)
$P_{i,t}^{as}$: Available photovoltaic power output (kW)
$\overline{\lambda}_t^{em}/\underline{\lambda}_t^{em}$: Upper/lower bound of energy market price ($/kWh)
$\overline{\lambda}_t^{rum}/\underline{\lambda}_t^{rum}$: Upper/lower bound of regulation up market price ($/kWh)
$\overline{\lambda}_t^{rdm}/\underline{\lambda}_t^{rdm}$: Upper/lower bound of regulation down market price ($/kWh)
$\overline{\lambda}_t^{srm}/\underline{\lambda}_t^{srm}$: Upper/lower bound of spinning reserve market price ($/kWh)
$\overline{\lambda}_t^{nsrm}/\underline{\lambda}_t^{nsrm}$: Upper/lower bound of non-spinning reserve market price ($/kWh)
$\overline{P}_{i,t}^{aw}/\underline{P}_{i,t}^{aw}$: Upper/lower bound of available wind power output (kW)
$\overline{P}_{i,t}^{as}/\underline{P}_{i,t}^{as}$: Upper/lower bound of available photovoltaic power output (kW)
$\Gamma^m/\Gamma_{i,t}^w/\Gamma_{i,t}^s$: Control parameter of electricity price/ wind power output/photovoltaic power output

Variables

$P_{i,t}^{em}$: Power traded (sold if positive and purchased if negative) in energy market (kW)
$R_{i,t}^{ru/rd/sr/nsr}$: Capacity provided in regulation up/ regulation down/spinning reserve/non-spinning reserve market (kW)
$P_{i,t}^{gt}$: Electrical power production of gas turbine (kW)
$H_{i,t}^{hrs}$: Thermal power production of heat recovery system (kW)
$H_{i,t}^{gb}$: Thermal power production of gas boiler (kW)
$P_{i,t}^{w}/P_{i,t}^{s}$: Wind/photovoltaic power output (kW)
$P_{i,t}^{ec}/Q_{i,t}^{ec}$: Electrical power input and cooling power output of electric chiller (kW)
$H_{i,t}^{ac}/Q_{i,t}^{ac}$: Thermal power input and cooling power output of absorption chiller (kW)
P_i^{ce}/P_i^{de}: Charge/discharge electrical power of electrical energy storage (kW)
$S_{i,t}^{ees}$: Electrical energy stored in electrical energy storage (kWh)
H_i^{ct}/H_i^{dt}: Charge/discharge thermal power of thermal energy storage (kW)
$S_{i,t}^{tes}$: Thermal energy stored in thermal energy storage (kWh)
$H_{i,j,t}^{tf}/Q_{i,j,t}^{cf}$: Thermal/cooling power flow transferred from area i to area j (kW)
$\mu_{i,t}^{o/su/sd/sb}$: Binary variable denoting gas turbine status (1 for operation/startup/shutdown/standby, 0 for otherwise)